Problem-1: Projective measurements

1 Solution

ⓐ $P(k) = Tr[P_k \rho]$ – trace-probability relationship

$= Tr[P_k |\phi\rangle\langle\phi|]$

$= Tr[\langle\phi|P_k|\phi\rangle]$ – cyclic properties of trace

$= \langle\phi|P_k|\phi\rangle$

$Tr[ABC] = Tr[CAB] = Tr[BCA]$

Where: Trace of constant is itself

$P(k) = \langle\phi|P_k|\phi\rangle$

$Tr[C] = C$

$Tr[\langle\phi|P_k|\phi\rangle] = \langle\phi|P_k|\phi\rangle$

ⓑ $P(k) = \langle\phi|P_k|\phi\rangle$

$= \langle\phi|\left(\sum_{i=1}^{n_k} |\psi_i^k\rangle\langle\psi_i^k|\right)|\phi\rangle$

$= \sum_{i=1}^{n_k} \langle\phi|\psi_i^k\rangle\langle\psi_i^k|\phi\rangle$

$= \sum_{i=1}^{n_k} |\langle\phi|\psi_i^k\rangle|^2$

2. What is the state of the system after measuring the outcome λ_k?

Solution

After measurement the state of system $|\phi'\rangle$ is given by

$$|\phi'\rangle = \frac{P_k|\phi\rangle}{\sqrt{\langle\phi|P_k|\phi\rangle}}$$

$$= \frac{\sum_{i=1}^{n_k} |\psi_i^k\rangle\langle\psi_i^k|\phi\rangle}{\sqrt{\sum_{i=1}^{n_k} |\langle\psi_i^k|\phi\rangle|^2}} = \frac{\sum_{i=1}^{n_k}\left(|\psi_i^k\rangle\langle\psi_i^k|\right)|\phi\rangle}{\sqrt{\sum_{i=1}^{n_k} |\langle\psi_i^k|\phi\rangle|^2}} = \frac{P_k|\phi\rangle}{\sqrt{P(k)}}$$

1

3. Consider observable, $A = X \otimes X$

Given

$A = X \otimes X$

$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

Requires

- Spectral decomposition?
- Eigenvalues = ?
- Eigenvector = ?

Solution

1- Calculated eigenvalues and Eigenvectors of X

Eigenvalues (λ)

$X|\psi\rangle = \lambda|\psi\rangle$

$(X - \lambda I)|\psi\rangle = 0$

$\det(X - \lambda I) = 0$

$\det\left[\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}\right] = 0$

$\det\begin{bmatrix} -\lambda & 1 \\ 1 & -\lambda \end{bmatrix} = 0$

$(-\lambda)(-\lambda) - (1)(1) = 0$

$\lambda^2 - 1 = 0$

$\lambda = \pm 1$

$\left.\begin{array}{l} \lambda_1 = 1 \\ \lambda_2 = -1 \end{array}\right\}$

Calculates the eigen state

By using λ_1 and λ_2

For $\lambda_1 = 1$

$|\psi_1\rangle = \begin{pmatrix} a \\ b \end{pmatrix}$

$(X - \lambda I)|\psi_1\rangle = 0$

$\left(\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right)\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$\left.\begin{array}{l} -a + b = 0 \\ a - b = 0 \end{array}\right\} \underline{a = b}$

found the values of "a" and "b"

$|\psi\rangle = a|0\rangle + b|1\rangle$

$\langle\psi|\psi\rangle = 1$

$|a|^2 + |b|^2 = 1$

$|a|^2 + |a|^2 = 1 \}, a = b$

$2|a|^2 = 1$

$|a|^2 = \frac{1}{2}$

$|a| = \frac{1}{\sqrt{2}}$

$\left.\begin{array}{l} a = \frac{1}{\sqrt{2}} \\ b = \frac{1}{\sqrt{2}} \end{array}\right\}$

3- Spectral decomposition of $A = X \otimes X$

$A = X \otimes X$

$= (|+\rangle\langle+| - |-\rangle\langle-|) \otimes (|+\rangle\langle+| - |-\rangle\langle-|)$

$= \underbrace{|++\rangle\langle++|}_{P_{+1} = 1} - \underbrace{|+-\rangle\langle+-|}_{P_{-1} = -1} - \underbrace{|-+\rangle\langle-+|}_{P_{-1} = -1} + \underbrace{|--\rangle\langle--|}_{P_{+1} = 1}$

$= \underbrace{|++\rangle\langle++| + |--\rangle\langle--|}_{P_{+1} = 1} - \underbrace{(|+-\rangle\langle+-| + |-+\rangle\langle-+|)}_{P_{-1} = -1}$

The projectors are

$P_{+1} = |++\rangle\langle++| + |--\rangle\langle--|$

$P_{-1} = |+-\rangle\langle+-| + |-+\rangle\langle-+|$

$A = \sum_{i}^{nk} \lambda_k P_k$

$A = (+1) P_{+1} + (-1) P_{-1}$

$A = P_{+1} - P_{-1}$

$A = P_{+1} - P_{-1}$

4- Given

$|\phi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + i|11\rangle)$

$A = X \otimes X$

Required

$\langle A \rangle = ?$

Solution

$\langle A \rangle = \langle \phi | A | \phi \rangle$

$= \left[\frac{1}{\sqrt{2}}(\langle 00| - i\langle 11|) (X \otimes X) \frac{1}{\sqrt{2}}(|00\rangle + i|11\rangle) \right]$

Then

$|\psi_1\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$

$= \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$

$|\psi_1\rangle = |+\rangle$

For $\lambda_2 = -1$

$(X - \lambda_2 I)|\psi_2\rangle = 0$

$$\left[\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} - \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}\right]\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\left.\begin{array}{l} a+b=0 \\ a+b=0 \end{array}\right\} a = -b$$

$|\psi_2\rangle = a|0\rangle + b|1\rangle$

calculated the "a" and "b"

$\langle\psi_2|\psi_2\rangle = 1$

$|a|^2 + |b|^2 = 1$

$|a|^2 + |a|^2 = 1$

$2|a|^2 = 1$

$|a|^2 = \frac{1}{2}$

$|a| = \frac{1}{\sqrt{2}}$

$$\left.\begin{array}{l} a = \frac{1}{\sqrt{2}} \\ b = -\frac{1}{\sqrt{2}} \end{array}\right\}$$

$|\psi_2\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}} = |-\rangle$

There fore the eigen state and eigenvalue

$$\left.\begin{array}{l} \lambda_1 = 1 \\ \lambda_2 = -1 \end{array}\right\} \text{Eigenvalues}$$

$$\left.\begin{array}{l} |\psi_1\rangle = |+\rangle \\ |\psi_2\rangle = |-\rangle \end{array}\right\} \text{eigenstate}$$

2- Spectral decomposition of X

$$X = \sum_{i=1}^{2} \lambda_i |\psi_i\rangle\langle\psi_i|$$

$X = (+1)|+\rangle\langle+| + (-1)|-\rangle\langle-|$

$X = |+\rangle\langle+| - |-\rangle\langle-|$

$$\langle A \rangle = \frac{1}{2}\left(\langle 00| - i\langle 11|\right)\left(X|0\rangle \otimes X|0\rangle + i(X|1\rangle \otimes X|1\rangle)\right)$$

$$= \frac{1}{2}\left(\langle 00| - i\langle 11|\right)\left(|11\rangle + i|00\rangle\right)$$

$$= \frac{1}{2}\left(\underbrace{\langle 00|11\rangle}_{=0} + i\underbrace{\langle 00|00\rangle}_{=1} - i\underbrace{\langle 11|11\rangle}_{=1} - i\underbrace{\langle 11|00\rangle}_{=0}\right)$$

$$= \frac{1}{2}\left(0 + i - i + 0\right)$$

$$= \frac{0}{2}$$

$$\underline{\underline{\langle A \rangle = 0}}$$

Problem:2 POVMs

1) Given

$E_i > 0$

$\sum_i E_i = 1$

Required

$\left.\begin{array}{l} p_i \geq 0 \\ \sum p_i = 1 \end{array}\right\}$ show and explain this

Solution

1. Necessity for non-negativity ($p_i \geq 0$)

- probability must be $p_i \geq 0$ for all physical states ρ

$$p_i = Tr(\rho E_i)$$

- $E_i \geq 0$, $\rho \geq 0$ which implies that $p_i \geq 0$

$$\left.\begin{array}{l} E_i \geq 0 \\ \rho \geq 0 \end{array}\right\} \Rightarrow p_i \geq 0$$

2. Necessity for summing to one ($\sum_i p_i = 1$)

$$\sum_i p_i = 1$$

Summing the probabilities using Born Rule

$$p_i = Tr(\rho E_i) \Rightarrow \sum_i p_i = \sum_i Tr(\rho E_i)$$
$$= Tr(\rho \underbrace{\sum_i E_i}_{=1}) = Tr(\rho \sum_i E_i) = Tr(\rho)$$
$$= Tr(\rho)$$
$$\sum_i p_i = 1$$

Therefore the two are necessity

$$\left.\begin{array}{l} p_i \geq 0 \\ \sum_i p_i = 1 \end{array}\right\}$$

2. Show that every projective measurement is type of POVM

Solution

Projective measurement (PM)

A Set of operators $\{P_i\}$ constitutes a projective measurement If each P_i is a projection operator, satisfying

1. Hermiticity: $P = P^{+}$
2. projection operator $P_i^2 = P_i$
3. completeness $\sum_i P_i = I$

POVM

A set of operators $\{E_i\}$ is POVM if its elements satisfy

1. Positivity: $E_i \geq 0$
2. completeness: $\sum_i E_i = I$

Showing projective is POVM

1- completeness condition
2- positivity condition

} since the set of projective operator $\{P_i\}$ satisfies

$$P_i \geq 0$$

$$\sum_i P_i = 1$$

Therefore every projective measurement is type of POVM

(3). Why we not consider every POVM as projective measurement?

Solution

We cannot consider every POVM as projective measurement because a POVM E_k is only required to be positive operator

$$E_k \geq 0 \rightarrow \text{it is POVM}$$

While projective element P_i must be satisfying $P_i^2 = P_i$

Example

$$E_1 = \begin{pmatrix} 1/3 & 0 \\ 0 & 2/3 \end{pmatrix}, \quad E_2 = \begin{pmatrix} 2/3 & 0 \\ 0 & 1/3 \end{pmatrix}$$

1. POVM Check (Satisfied)

- Positivity, All eigenvalue $(1/3, 2/3)$ $E_1, E_2 \geq 0$
- Completeness $I = E_1 + E_2 = \begin{pmatrix} 1/3 & 0 \\ 0 & 2/3 \end{pmatrix} + \begin{pmatrix} 2/3 & 0 \\ 0 & 1/3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

2. Projective measurement (Failed)

$$E_1^2 \neq E_1$$

$$E_1^2 = \begin{pmatrix} 1/3 & 0 \\ 0 & 2/3 \end{pmatrix}^2 = \begin{pmatrix} 1/9 & 0 \\ 0 & 4/9 \end{pmatrix}$$

$$\therefore E_1^2 \neq E_1$$

} Therefore the set $\{E_1, E_2\}$ is POVM but not projective measurement because of $E_1^2 \neq E_1$

Problem 3. Density Matrices

(1) Given

$$\rho = \sum_i p_i \left(|a_i\rangle \otimes |B_i\rangle\right)\left(\langle a_i| \otimes \langle B_i|\right)$$

Required

$\left.\begin{matrix}\rho_1\\ \rho_2\end{matrix}\right\}$ Show that it's indeed separable

Solution

$$\rho_1 = \frac{1}{2}\left(|\phi^+\rangle\langle\phi^+| + |\phi^-\rangle\langle\phi^-|\right) \qquad |\phi^\pm\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle \pm |11\rangle\right)$$

$$\rho_2 = \frac{1}{2}\left(|\psi^+\rangle\langle\psi^+| + |\psi^-\rangle\langle\psi^-|\right) \qquad |\psi^\pm\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle \pm |10\rangle\right)$$

Express the state in the computation basis both ρ_1 and ρ_2

$$\rho_1 = \frac{1}{2}\left(\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)\cdot\frac{1}{\sqrt{2}}(\langle 00| + \langle 11|)\right) + \frac{1}{2}\left(\frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)\frac{1}{\sqrt{2}}(\langle 00| - \langle 11|)\right)$$

$$= \frac{1}{4}\Big[|00\rangle\langle 00| + \underbrace{|00\rangle\langle 11|}_{x} + \underbrace{|11\rangle\langle 00|}_{xx} + |11\rangle\langle 11| + |00\rangle\langle 00| - \underbrace{|00\rangle\langle 11|}_{x} - \underbrace{|11\rangle\langle 00|}_{xx} + |11\rangle\langle 11|\Big]$$

$$= \frac{1}{4}\left[2|00\rangle\langle 00| + 2|11\rangle\langle 11|\right] = \frac{2}{4}\left[|00\rangle\langle 00| + |11\rangle\langle 11|\right]$$

$$\rho_1 = \frac{1}{2}\left[|00\rangle\langle 00| + |11\rangle\langle 11|\right]$$

$$\rho_2 = \frac{1}{2}\left(|\psi^+\rangle\langle\psi^+| + |\psi^-\rangle\langle\psi^-|\right)$$

$$= \frac{1}{2}\left[\frac{1}{2}(|01\rangle + |10\rangle)(\langle 01| + \langle 10|) + \frac{1}{2}(|01\rangle - |10\rangle)(\langle 01| - \langle 10|)\right]$$

$$= \frac{1}{4}\Big[|01\rangle\langle 01| + \underbrace{|01\rangle\langle 10|}_{x} + \underbrace{|10\rangle\langle 01|}_{xx} + |10\rangle\langle 10| + |01\rangle\langle 01| - \underbrace{|01\rangle\langle 10|}_{x} - \underbrace{|10\rangle\langle 01|}_{xx} + |10\rangle\langle 10|\Big]$$

$$= \frac{1}{4}\left[2|01\rangle\langle 01| + 2|10\rangle\langle 10|\right] = \frac{2}{4}\left[|01\rangle\langle 01| + |10\rangle\langle 10|\right]$$

$$= \frac{1}{2}\left(|01\rangle\langle 01| + |10\rangle\langle 10|\right)$$

The result are mixture of the product state

$$\rho_1 = \frac{1}{2}(|00\rangle\langle 00| + |11\rangle\langle 11|)$$

with

$p_1 = \frac{1}{2}, |a_1\rangle = |0\rangle, |b_1\rangle = |0\rangle$

$p_2 = \frac{1}{2}, |a_2\rangle = |1\rangle, |b_2\rangle = |1\rangle$

So the ρ_1 can be written as sum of product state, it is separable

$$\rho_2 = \frac{1}{2}(|01\rangle\langle 01| + |10\rangle\langle 10|)$$

with

$p_1 = \frac{1}{2}, |a_1\rangle = |0\rangle, |b_1\rangle = |1\rangle$

$p_2 = \frac{1}{2}, |a_2\rangle = |1\rangle, |b_2\rangle = |0\rangle$

So the ρ_2 can be written as sum of product state, it is separable

(2.1) Given

$\rho_{AB} = \rho_A \otimes \rho_B$

Required

Prove

ρ_{AB} Separable $\Rightarrow$ Purity(ρ_{AB}) $\leq$ Purity(ρ_A)

$Tr(\rho^2_{AB}) \leq Tr(\rho_A^2)$

Solution

Separable state

$$\rho_{AB} = \sum_i p_i \rho_A^{(i)} \otimes \rho_B^{(i)}, \quad p_i \geq 0, \sum p_i = 1$$

Compute the purity of the joint state

$$Tr(\rho_{AB}^2) = Tr\left(\sum_{i,j} p_i p_j \rho_A^{(i)} \rho_A^{(j)} \otimes \rho_B^{(i)} \rho_B^{(j)}\right)$$

$$= \sum_{i,j} p_i p_j \, Tr(\rho_A^{(i)} \rho_A^{(j)}) \, Tr(\rho_B^{(i)} \rho_B^{(j)})$$

The reduced state A is $\rho_A = \sum_i p_i \rho_A^{(i)}$ so the purity

$$Tr(\rho_A^2) = \sum_{i,j} p_i p_j \, Tr(\rho_A^{(i)} \rho_A^{(j)})$$

$$=$$

Apply the inequality

The Cauchy - Schwarz inequality for product state

Let say, two density matrices

(σ, τ)

$$|Tr(\sigma\tau)|^2 \le Tr(\sigma^2)\, Tr(\tau^2)$$

so

$$Tr(\sigma\tau) \le \sqrt{Tr(\sigma^2)\, Tr(\tau^2)}$$

where

$$Tr(\sigma^2) \le 1$$

$$Tr(\sigma\tau) \le \sqrt{Tr(\sigma^2)\, Tr(\tau^2)} \le 1.$$

Applying on the proof

$Tr(\rho_B^{(i)} \rho_B^{(j)})$ is just of this form with

$$\sigma = \rho_B^{(i)}, \quad \tau = \rho_B^{(j)}$$

$$Tr(\rho_B^{(i)} \rho_B^{(j)}) \le 1$$

+ The double sum

$$Tr(\rho_{AB}^2) = \sum_{ij} p_i p_j \, Tr(\rho_A^{(i)} \rho_A^{(j)})\, Tr(\rho_B^{(i)} \rho_B^{(j)})$$

Each factor $\mathrm{Tr}(\rho_B^{(i)} \rho_B^{(j)})$ multiplies a nonnegative term

$p_i p_j \mathrm{Tr}(\rho_A^{(i)} \rho_A^{(j)})$

Since $\mathrm{Tr}(\rho_A^{(i)} \rho_B^{(j)}) \leq 1$ every term in the sum is less than or equal to the same term if we replaced that by factor by 1

$$\mathrm{Tr}(\rho^2_{AB}) \leq \sum_{ij} p_i p_j \mathrm{Tr}(\rho_A^{(i)} \rho_A^{(j)})$$

$$\mathrm{Tr}(\rho^2_{AB}) \leq \mathrm{Tr}(\rho_A^2)$$

$$\mathrm{Tr}(\rho^2_{AB}) \leq \mathrm{Tr}(\rho^2_A)$$

(3.) Consider a two-qubit system $\mathcal{H} = \mathbb{C}^2 \otimes \mathbb{C}^2$

$$\rho = \frac{1}{2}\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

(a) Is the state ρ a pure or a mixed state? Prove your statement.

Solution

Density matrix ρ is pure state it and only if $\rho^2 = \rho$ otherwise it mixed

$$\rho^2 = \rho \Rightarrow Tr(\rho^2) = Tr(\rho) = 1$$

$$\boxed{\begin{array}{l} Tr(\rho^2) = 1 \rightarrow \text{pure state} \\ Tr(\rho^2) < 1 \rightarrow \text{mixed stat} \end{array}}$$

calculate ρ^2

$$\rho^2 = \frac{1}{2}\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \frac{1}{2}\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = \frac{1}{4}\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 2 & 2 & 0 \\ 0 & 2 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$= \frac{2}{4}\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

calculates purity

$$Tr(\rho^2) = Tr\left(\frac{1}{2}\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}\right) = \frac{1}{2}(1+1) = \frac{2}{2}$$

$Tr(\rho^2) = 1$, therefore the state is Pure State

(b) Is the state ρ a separable or Entangled state?

Solution

Since the state ρ is a pure state ~~[illegible]~~ we use the coefficient matrix C

P.T.O

Simple product of two single-qubit state,

$$|\psi\rangle = |\psi\rangle_A \otimes |\psi\rangle_B$$

Generally

$$|\psi\rangle = c_{00}|00\rangle + c_{01}|01\rangle + c_{10}|10\rangle + c_{11}|11\rangle$$

The coefficient matrix C for 2x2 matrix

$$C = \begin{pmatrix} c_{00} & c_{01} \\ c_{10} & c_{11} \end{pmatrix}, \quad \begin{array}{l} \det(C) = 0, \text{ separable} \\ \det(C) \neq 0, \text{ Entangled} \end{array}$$

Applying the state Test

Identify the state vector and coefficient

$$|\psi\rangle = \frac{1}{\sqrt{2}}|01\rangle + \frac{1}{\sqrt{2}}|10\rangle$$

The coefficient are

$$c_{00} = 0, \; c_{01} = \frac{1}{\sqrt{2}}, \; c_{10} = \frac{1}{\sqrt{2}}, \; c_{11} = 0$$

Coefficient matrix C

$$C = \begin{pmatrix} c_{00} & c_{01} \\ c_{10} & c_{11} \end{pmatrix} = \begin{pmatrix} 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 0 \end{pmatrix}$$

Simple it is Entangle

Because it has
$\underbrace{\text{Rank}(C) = 2}_{\text{Entangled}}$.
$\underbrace{\text{Rank}(C) = 1}_{\text{separable}}$

Calculate the determinant

$$\det(C) = \det\begin{pmatrix} 0 & \frac{1}{2} \\ \frac{1}{2} & 0 \end{pmatrix} = 0.0 - \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}}$$

$\det(C) = -\frac{1}{2}$, The density state Entangled state

Because $\det(C) \neq 0$, which is Entangled.